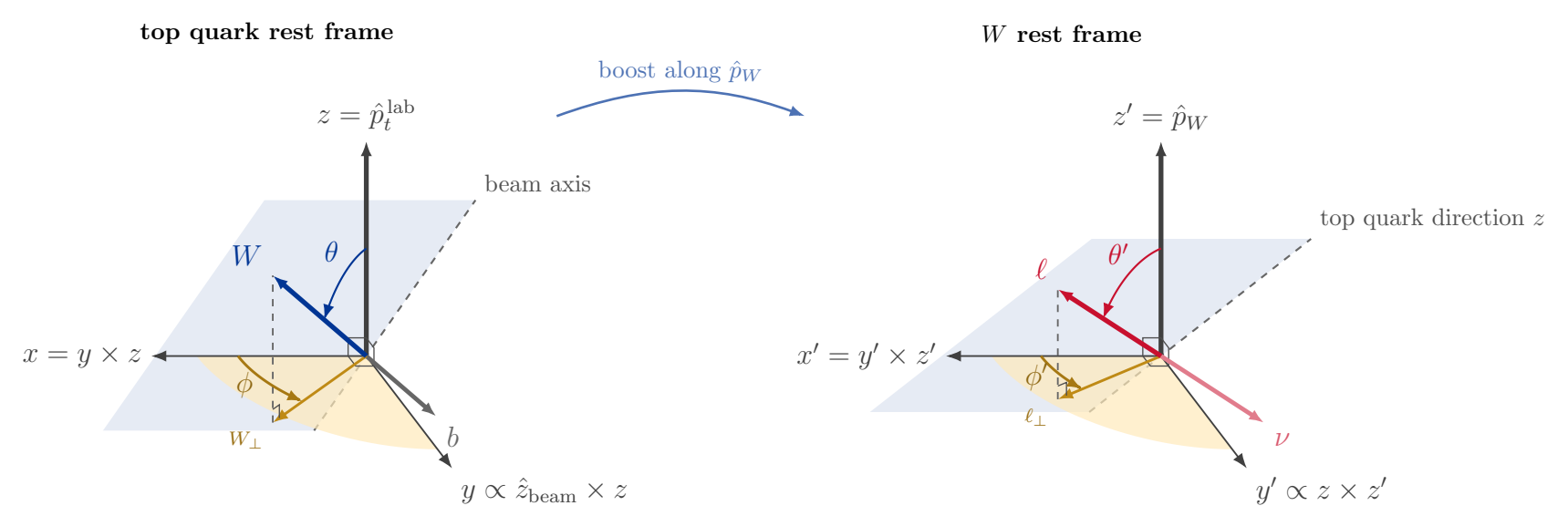

top quark rest frame
boost along $\hat{p}_W$
W rest frame
$z = \hat{p}_t^{\text{lab}}$
beam axis
W
θ
$x = y \times z$
ϕ
$W_\perp$
b
$y \propto \hat{z}_{\text{beam}} \times z$
$z' = \hat{p}_W$
top quark direction z
ℓ
θ'
$x' = y' \times z'$
ϕ'
$\ell_\perp$
ν
$y' \propto z \times z'$